



Stony Brook Institute
at Anhui University

安徽大学纽约石溪学院

ISE 333: User Interface Development (Design)

Chapter 2: Universal Usability

Spring 2026

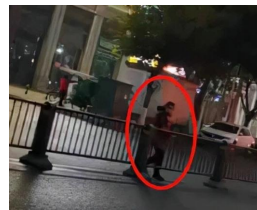
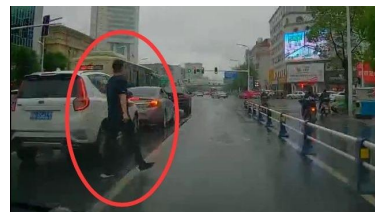
Instructor: Hao Li

Introduction

Just like an experienced driver who should try best to handle all these as good as possible

- Human Diversity

- The remarkable diversity of human abilities, backgrounds, motivations, personalities, cultures, and work styles challenges UIDers
 - The huge international consumer market in mobile devices has raised the pressure for UI designs that are universally usable
- Rethinking UI designs for differing situations often results in a better product for all users
 - Measures to accommodate special needs of one group (e.g. sidewalk curb cuts for wheelchair users), often have payoffs for many groups (e.g. baby strollers, wheeled luggage, handcarts)
- As a profession, UIDers will be remembered for how well they meet their users' needs. That is the ultimate goal: addressing the needs of all users

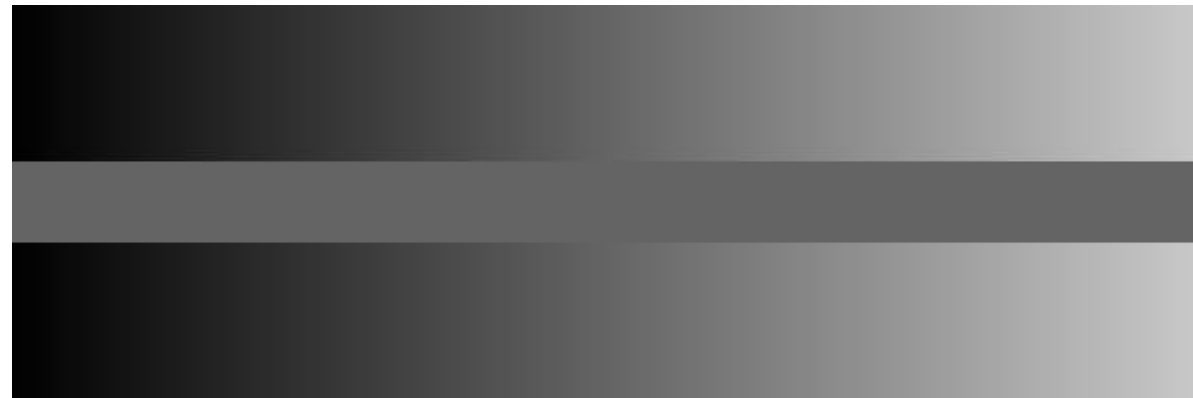
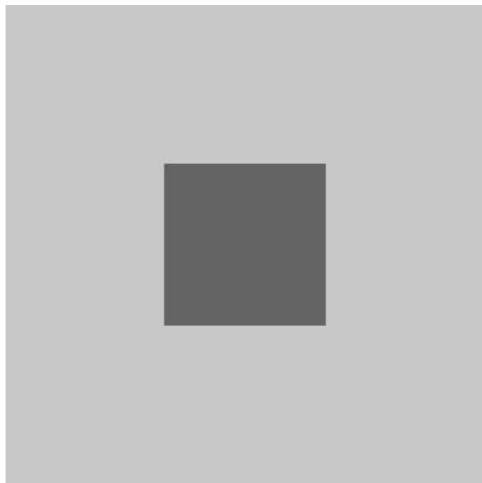
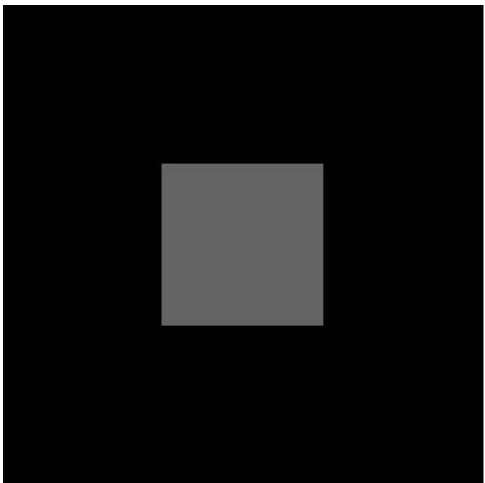


Individual Diversity

- Variations in Physical Abilities
 - Data statistics accumulated by ergonomics researchers and practitioners
 - Substantial experience is gained from design projects with automobiles, aircraft, cellphones, and so on
 - This design can be applied to UID
 - Data statistics about human dimensions
 - Thousands of measures of hundreds of features of people --- male and female, young and adult, Asian and European, underweight and overweight, tall and short --- provide data to construct 5- to 95-percentile design ranges
 - Great diversity in these *static measures* reminds us that there can be no image of an “average ” user and that compromises must be made or multiple versions of a system must be constructed
 - Measures of dynamic actions, e.g. reach distance while seated, speed of finger presses, or strength of lifting, are also necessary

Individual Diversity

- Variations in Physical Abilities
 - Since so much of work is related to **perception**, UIDers need to be aware of the ranges of human perceptual abilities, especially with regard to vision
 - Perception can be mysterious
 - The visual system responds differently to various colors
 - People's spectral range and sensitivity vary
 - Peripheral vision is quite different from the perception of images in the fovea (the central part of the retina)



Individual Diversity

- Variations in Physical Abilities
 - Since so much of work is related to **perception**, UIDers need to be aware of the ranges of human perceptual abilities, especially with regard to vision
 - Perception can be mysterious
 - The visual system responds differently to various colors
 - People's spectral range and sensitivity vary
 - Peripheral vision is quite different from the perception of images in the fovea (the central part of the retina)
 - Other senses are also important
 - Touch for keyboard or touch-screen
 - Hearing for audible cues
 - Speech input/output

Individual Diversity

- Variations in Physical Workplaces
 - Physical abilities also play a prominent role in the design of workplaces or workstations
 - The *Human Factors Engineering of Computer Workstations* standard lists these concerns
 - Worktable and display support height
 - Clearance under work surface for legs
 - Work-surface width and depth
 - Adjustability of heights and angles for chairs and work surfaces
 - Posture --- seating depth and angle, backrest height, etc.
 - Availability of armrests, footrests, and palmrests
 - Use of chair casters

一屋不扫何以扫天下

External ritual catalyses internal sublimation

Individual Diversity

- Variations in Physical Workplaces
 - Physical abilities also play a prominent role in the design of workplaces or workstations
 - The *Human Factors Engineering of Computer Workstations* standard lists these concerns
 - Workplace design is important in ensuring high job satisfaction, good performance, and low error rates
 - Designs can be compromised by a noisy environment, poor lighting, or a stuffy room, and that compromise will eventually lower performance, raise error rates, and discourage even motivated users
 - Another physical-environment consideration involves room layout and the sociology of human interaction

Individual Diversity

- Variations in Physical Abilities and Physical Workplaces
- Diverse Cognitive and Perceptual Abilities
 - A vital foundation for interactive-system designers is an understanding of the cognitive and perceptual abilities of the users
 - Users' cognitive and perceptual abilities can be very different
 - The perceptual abilities here actually refer to **high/cognitive-level perceptual abilities**, in contrast with low/physical-level perceptual abilities mentioned previously



Cognitive-level perception $\neq$ Good sight

Individual Diversity

- Variations in Physical Abilities and Physical Workplaces
- Diverse Cognitive and Perceptual Abilities
 - A vital foundation for interactive-system designers is an understanding of the cognitive and perceptual abilities of the users
 - Classification of human cognitive processes
 - Short-term and working memory
 - Long-term and semantic memory
 - Problem solving and reasoning
 - Decision making and risk assessment
 - Language communication and comprehension
 - Search, imagery, and sensory memory
 - Learning, skill development, knowledge acquisition, and concept attainment

Individual Diversity

- Variations in Physical Abilities and Physical Workplaces
- Diverse Cognitive and Perceptual Abilities
- Personality Differences
 - Phenomena of personality differences with respect to UIs
 - Some people are eager to use computers and mobile devices, while others find them frustrating
 - Even people who enjoy using these technologies may have very different preferences for interaction styles, pace of interaction, graphics versus tabular presentations, dense versus sparse data presentation, etc.
 - A clear understanding of personality and cognitive styles can be helpful in UID for diverse communities of users
 - Unfortunately there is **no simple taxonomy** of user personality types

Individual Diversity

- Personality Differences

- Big Five Test, a popular yet controversial technique

- Openness to experience/Intellect (closed/open) 开放性
 - Conscientiousness (disorganized/organized) 自律性
 - Extraversion (introverted/extroverted) 外向性
 - Agreeableness (disagreeable/agreeable) 温润性
 - Neuroticism (calm/nervous) 稳重性

- Study user behaviour for personality assessment

(官人性)

1. Is outgoing, sociable	Strongly Disagree	1	2	3	4	5	Strongly Agree
2. Is compassionate, has a soft heart	Strongly Disagree	1	2	3	4	5	Strongly Agree
3. Tends to be disorganized	Strongly Disagree	1	2	3	4	5	Strongly Agree
4. Is relaxed, handles stress well	Strongly Disagree	1	2	3	4	5	Strongly Agree
5. Has few artistic interests	Strongly Disagree	1	2	3	4	5	Strongly Agree
6. Has an assertive personality	Strongly Disagree	1	2	3	4	5	Strongly Agree
7. Is respectful, treats others with respect	Strongly Disagree	1	2	3	4	5	Strongly Agree
8. Tends to be lazy	Strongly Disagree	1	2	3	4	5	Strongly Agree
9. Stays optimistic after experiencing a setback	Strongly Disagree	1	2	3	4	5	Strongly Agree
10. Is curious about many different things	Strongly Disagree	1	2	3	4	5	Strongly Agree
11. Rarely feels excited or eager	Strongly Disagree	1	2	3	4	5	Strongly Agree

听其言观其行

Observe-Think-Reflect by heart

Universal Usability

- Practice
 - Which of the following is not related to individual diversity?
 - [1] Variations in physical abilities; [2] Diverse cognitive abilities; [3] Cultural differences; [4] Personality differences;
 - There is a commonly-agreed classification of user personality types, T/F?
 - The Big Five Test is not the only method to do personality assessment, T/F?
 - What is the difference between physical-level perception and cognitive-level perception?
 - Why do UIDers need so many experience/case/induction-based guidelines (that you will learn in Chapter 3) instead of enjoying some generally-applicable UID formulas?
 - Use learned knowledge to explain why shape-distinction based (SDB) UIs are normally more preferred than light-distinction based (LDB) UIs?
 - But why are traffic lights LDB instead of SDB?

DISCUSSION

Q & A

Group/Community Diversity

- Cultural and International Diversity
 - Another perspective on user differences has to do with cultural, ethnic, racial, or linguistic background



European/USA-style
(aesthetic) preference



Chinese-style
(aesthetic) preference

Group/Community Diversity

- Cultural and International Diversity
 - Another perspective on user differences has to do with cultural, ethnic, racial, or linguistic background



Chinese-style colorful dressing



French/Italian-style black dressing

Group/Community Diversity

- Cultural and International Diversity
 - Another perspective on user differences has to do with cultural, ethnic, racial, or linguistic background
 - More and more is being learned about computer users from different cultures, but UIDers are still struggling to *establish guidelines* that are appropriate across multiple languages and cultures
 - Software architectures that facilitate customization of local versions of UIs offer a competitive advantage
 - Hardware issues include character sets, keyboards, and special input devices



En keyboard
(left)
vs.
(right)
Fr keyboard



Group/Community Diversity

- Cultural and International Diversity
 - UID concerns for internationalization include the following
 - Characters, numerals, special characters, and diacritics (e.g. beaut^é变音)
 - Left-to-right vs. right-to-left vs. vertical input and reading
 - Date and time formats
 - Numeric and currency formats
 - Weights and measures
 - Telephone numbers and addresses
 - Names and titles (Mr., Ms., Mme., M., Dr.)
 - Social security, national identification, and passport numbers
 - Capitalization and punctuation
 - Sorting sequences
 - Icons, buttons, and colours
 - Pluralization, grammar, and spelling
 - Etiquette, policies, tone, formality, and metaphors

Group/Community Diversity

- Cultural and International Diversity
 - UID concerns for internationalization include the following
 - Characters, numerals, special characters, and diacritics (e.g. beaut^é变音)
 -
 - Etiquette, policies, tone, formality, and metaphors (The list is long yet incomplete)
 - Recent studies of consumer use show performance and preference differences for information density, animation, cute characters, eagerness for timely updates, incentives for social participation, game-like features, etc.
 - Current highly competitive atmosphere means that more effective localization may produce a strong advantage
 - For effective UID, companies run *usability studies* with users from different countries, cultures, and language communities

Group/Community Diversity

- Cultural and International Diversity
- Users with Disabilities
 - When digital content and services can be flexibly presented in different formats, all users benefit
 - Flexibility is most appreciated by users with disabilities who now can access content and services using diverse input and output devices
 - Blind users may utilize screen readers or refreshable braille displays
 - Users with hearing impairments may need captioning on videos and transcripts of audio
 - Users with limited dexterity or other motor impairments may utilize speech recognition, eye-tracking, alternative keyboards or pointing devices
 - For UIs to be accessible for people with disabilities, they generally need to follow a set of UID *guidelines* for accessibility
 - *Web Content Accessibility Guidelines* (WCAG)

Group/Community Diversity

- Users with Disabilities
 - Web Content Accessibility Guidelines (WCAG)
 - <https://www.w3.org/TR/WCAG21/>
 - Basis for laws and regulations

TABLE OF CONTENTS	
	Abstract
	Status of This Document
	Introduction
	Background on WCAG 2
	WCAG 2 Layers of Guidance
	WCAG 2.1 Supporting Documents
	Requirements for WCAG 2.1
	Comparison with WCAG 2.0
	New Features in WCAG 2.1
	Numbering in WCAG 2.1
	Conformance to WCAG 2.1
	Later Versions of Accessibility Guidelines
1.	Perceivable
1.1	Text Alternatives
1.1.1	Non-text Content
1.2	Time-based Media
1.2.1	Audio-only and Video-only (Prerecorded)
1.2.2	Captions (Prerecorded)
1.2.3	Audio Description or Media Alternative (Prerecorded)
1.2.4	Captions (Live)
1.2.5	Audio Description (Prerecorded)
1.2.6	Sign Language (Prerecorded)
1.2.7	Extended Audio Description

Abstract

Web Content Accessibility Guidelines (WCAG) 2.1 covers a wide range of recommendations for making web content more accessible. Following these guidelines will make content more accessible to a wider range of people with disabilities, including accommodations for blindness and low vision, deafness and hearing loss, limited movement, speech disabilities, photosensitivity, and combinations of these, and some accommodation for learning disabilities and cognitive limitations; but will not address every user need for people with these disabilities. These guidelines address accessibility of web content on any kind of device (including desktops, laptops, kiosks, and mobile devices). Following these guidelines will also often make web content more usable to users in general.

WCAG 2.1 success criteria are written as testable statements that are not technology-specific. Guidance about satisfying the success criteria in specific technologies, as well as general information about interpreting the success criteria, is provided in separate documents. See [Web Content Accessibility Guidelines \(WCAG\) Overview](#) for an introduction and links to WCAG technical and educational material.

WCAG 2.1 extends [Web Content Accessibility Guidelines 2.0 \[WCAG20\]](#), which was published as a W3C Recommendation December 2008. Content that conforms to WCAG 2.1 also conforms to WCAG 2.0. The WG intends that for policies requiring conformance to WCAG 2.0, WCAG 2.1 can provide an alternate means of conformance. The publication of WCAG 2.1 does not deprecate or supersede WCAG 2.0. While WCAG 2.0 remains a W3C Recommendation, the W3C advises the use of WCAG 2.1 to maximize future applicability of accessibility efforts. The W3C also encourages use of the most current version of WCAG when developing or updating web accessibility policies.

Status of This Document

This section describes the status of this document at the time of its publication. A list of current W3C publications and the latest revision of this technical report can be found in the [W3C standards and drafts index](#) at <https://www.w3.org/TR/>.

Group/Community Diversity

- Cultural and International Diversity
- Users with Disabilities
- Older Users
 - Seniority offers many pleasures and all the benefits of experience, but aging can also have negative physical, cognitive, and social consequences.
Understanding the human factors of aging can help UIDers to create UIs that facilitate access by older users
 - A nonuniform set of progressive changes in physiological and psychological functioning ... Average visual and auditory acuity decline considerably with age, as do average strength and speed of response ... [People experience] loss of at least some kinds of memory function, decline in perceptual flexibility, slowing of “stimulus encoding”, and increased difficulty in the acquisition of complex mental skills ... Visual functions such as static visual acuity, dark adaptation, contrast sensitivity, and peripheral vision decline, on average, with age

Group/Community Diversity

UIDers



Genuine user needs

- Older Users
 - Despite above discouraging side, UIDers can do much to accommodate older users. As the world's population ages, UIDers in many fields are adapting their work to serve older users, which can benefit all users
 - Larger street signs, brighter traffic lights, and better nighttime lighting
 - Desktop, web, and mobile devices provide users with control over font sizes, display contrast, and audio levels
 - Easier-to-use pointing devices, clearer navigation paths, and consistent layouts
 - Considering older and disabled users during the UID process often produces novel solutions
 - Ballpoint pens (for people with impaired dexterity)
 - Cassette tape recorders (for blind users to listen to stories)
 - Auto-completion/suggesting software (to ease data input for older and disabled users)

穷则变
变则通
通则久

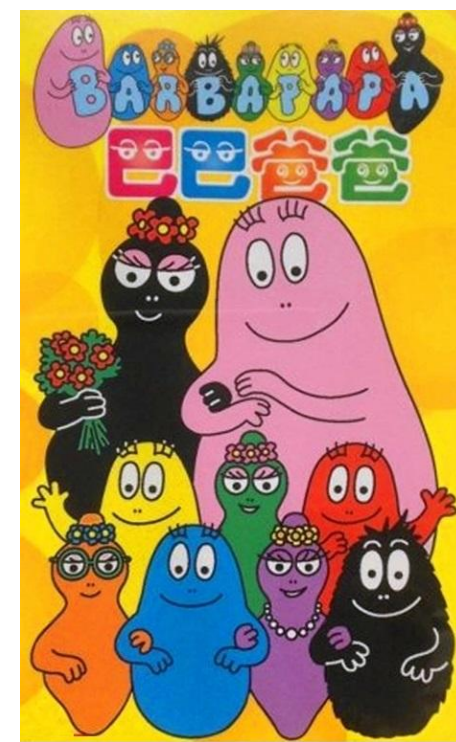
Since these things are (obviously) good to general users, why did not we do them at the very beginning?
Why was it after consideration of older/disabled users that we began to do these things and hence benefit all users?

Group/Community Diversity

- Older Users
 - Despite above discouraging side, UIDers can do much to accommodate older users. As the world's population ages, UIDers in many fields are adapting their work to serve older users, which can benefit all users
 - Considering older and disabled users during the UID process often produces novel solutions
 - Reports show that for older users who have fear of computers and doubt their ability to use computers, these fears tend to give way quickly after a few positive experiences
 - The older users feel satisfied after the positive experiences and are eager to learn more
- Summary for Older Users
 - Making UIs more attractive and accessible to older users enables them to take advantage of technology, enables others to benefit from their participation, and can make technology easier to everyone

Group/Community Diversity

- Cultural and International Diversity
- Users with Disabilities
- Older Users
- Children
 - Appropriate UID principles for children recognize young users' intense desire for the kind of interactive engagement that gives them control with appropriate feedback and supports their social engagement with peers
 - Children like familiar characters, exploratory environments, and the capacity for repetition. Younger children will sometimes replay a game, reread a story, or replay a music sequence dozens of times
 - While too much “screen time” can interfere with childhood development, well-designed applications can help children with physical, relationship, and emotional problems



Group/Community Diversity

- Children
 - Designing for younger children requires attention to their limitations
 - Their evolving dexterity means that mouse dragging, double-clicking, and small targets cannot always be used
 - Their low capacity for abstraction means that complex sequences must be avoided unless an adult is involved
 - Their short attention spans
 - Their limited capacity to work with multiple concepts
 - UIs for children should have a responsibility to attend to dangers, especially in web-based environments
 - Appropriate information for the education of children about privacy issues and threats from strangers is also a requirement
 - The capacity for playful creativity in art, music, and writing and the value of educational activities remain potent reasons to pursue good UIs for children

Universal Usability

- Individual Diversity
- Group/Community Diversity
- Hardware and Software Diversity
 - In addition to accommodating different classes of users and skill levels, UIDers need to support a wide range of hardware and software platforms
 - Despite rapid progress of hardware technology, UIDers need to accommodate older devices and deal with newer mobile devices that may have low-bandwidth connections and small screens
 - The challenge of accommodating diverse hardware is coupled with the need to ensure access through many generations of software. New operating systems, web browsers, e-mail clients, and application programs should provide backward compatibility in terms of their UID and file structures

Universal Usability

- Practice
 - Which of the following is not related to group/community diversity?
 - [1] Cultural conventions and norms; [2] Older and disabled users; [3] Individual personality and habits; [4] Children especially younger children;
 - UIDers should care only the cutting-edge hardware/software technology, T/F?
 - Software (e.g. OS) backward compatibility can be regarded as a UU goal, T/F?
 - Above UU goal (and other UU goals) are incorporated into previously-learned usability goals to form a larger set of goals that need trade-offs, any example?
 - UIDer A releases an educational solution by the pure text prompting way, while UIDer B releases one that provides the user an immersion in the story context of Barbapapa (children-favorite characters). Which tends to be more appropriate to children? Which tends to better suit adults? Why?
 - Knowledge points learned by so far seem to be management-like. Can you imagine a way of how they can be incorporated into a loop of engineering?

DISCUSSION

Q & A